



Machine Learning - Introduction

Agenda

- ❑ Machine Learning – Introduction
- ❑ Machine Learning – Applications
- ❑ Machine Learning – Types
- ❑ Machine Learning – Supervised Learning
- ❑ Machine Learning – UnSupervised Learning
- ❑ Machine Learning - Reinforcement learning

Machine Learning - Introduction

What is Machine Learning?

- Making predictions or decisions from data
- “Build a model that is a good and useful approximation to the data.”
- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” - Tom Mitchell

Machine Learning - Introduction

What famous people say about Machine Learning?

- “A breakthrough in machine learning would be worth ten Microsofts” - Bill Gates
- “Machine learning is the next Internet”
 - Tony Tether, Director, DARPA
- “Machine learning is today’s discontinuity”
 - Jerry Yang, ex-CEO, Yahoo

Machine Learning - Introduction

Tradition Programming



Machine Learning



Machine Learning - Introduction

Machine Learning

- **Seeds** = Algorithms
- **Nutrients** = Data
- **Gardener** = You
- **Plants** = Programs



Machine Learning – Related Terms

Machine Learning, Data Mining, Knowledge Discovery,
Artificial Intelligence, Statistical Learning, Pattern Recognition,
Computational Learning



Machine Learning – When to use?

- Human expertise does not exist
E.g. navigating on Mars
- Humans are unable to explain their expertise
E.g. speech recognition
- Solution changes in time
E.g. routing on a computer network
- Data is cheap and abundant; knowledge is expensive and scarce

Machine Learning – Applications

- Science (Astronomy, neuroscience, medical imaging, bio-informatics)
- Environment (energy, climate, weather, resources)
- Retail (Intelligent stock control, demographic store placement)
- Manufacturing (Intelligent control, automated)
- MonitoringSecurity (Intelligent smoke alarms, fraud detection)
- Marketing (promotions, ...)
- Management (Scheduling, timetabling)
- Finance (credit scoring, risk analysis...)
- Web data (information retrieval, information extraction, ...)

Machine Learning - Types

Machine Learning – Types:

- **Supervised learning:** Predict an output y when given an input x
 - ❖ Classification: if y is categorical
 - ❖ Regression: if y is continuous
- **Unsupervised learning:** Create an internal representation of the input, e.g. clustering, dimensionality
- Unsupervised learning is important in machine learning as getting labels is often difficult and expensive

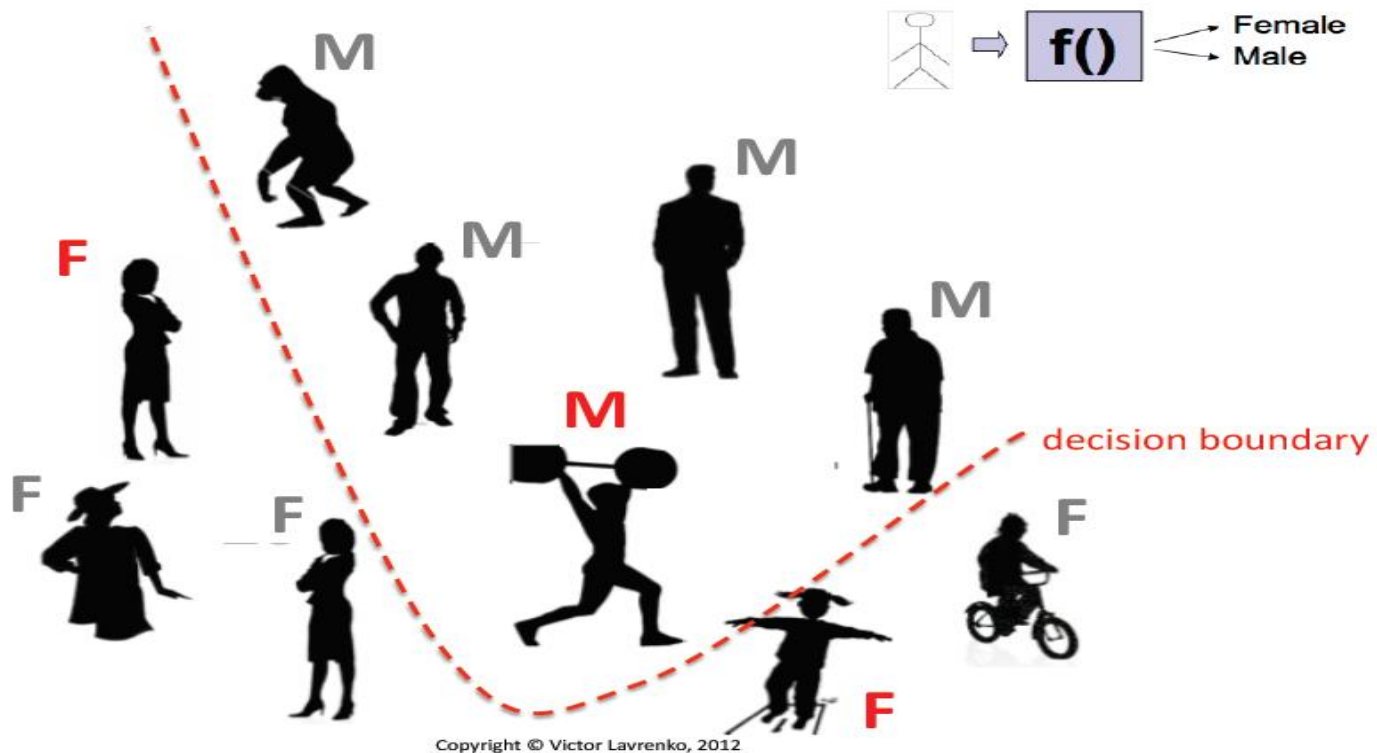
Machine Learning - Types

Machine Learning - Types

	<i>Supervised Learning</i>	<i>Unsupervised Learning</i>
<i>Discrete</i>	classification or categorization	clustering
<i>Continuous</i>	regression	dimensionality reduction

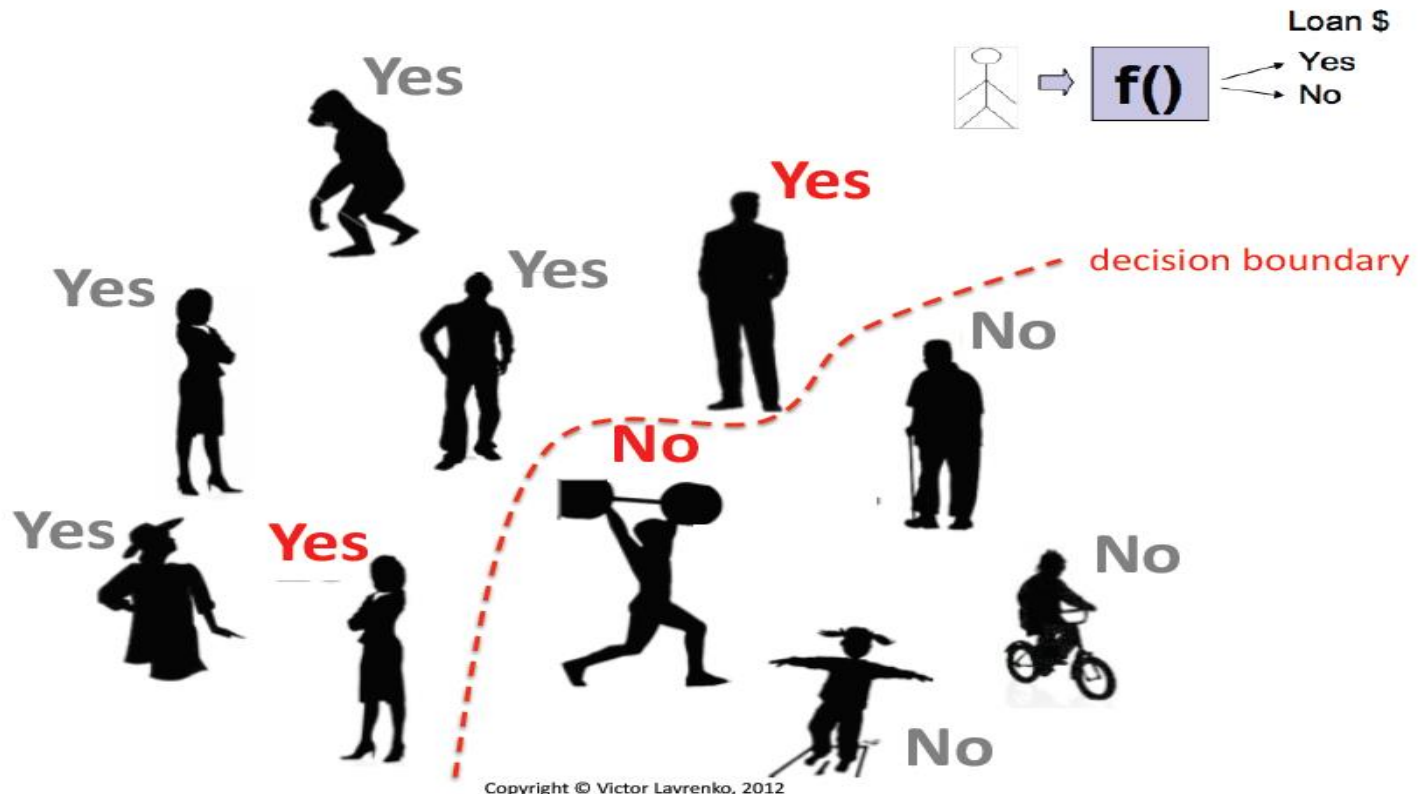
Machine Learning – Supervised Learning

Machine Learning – Supervised - Classification



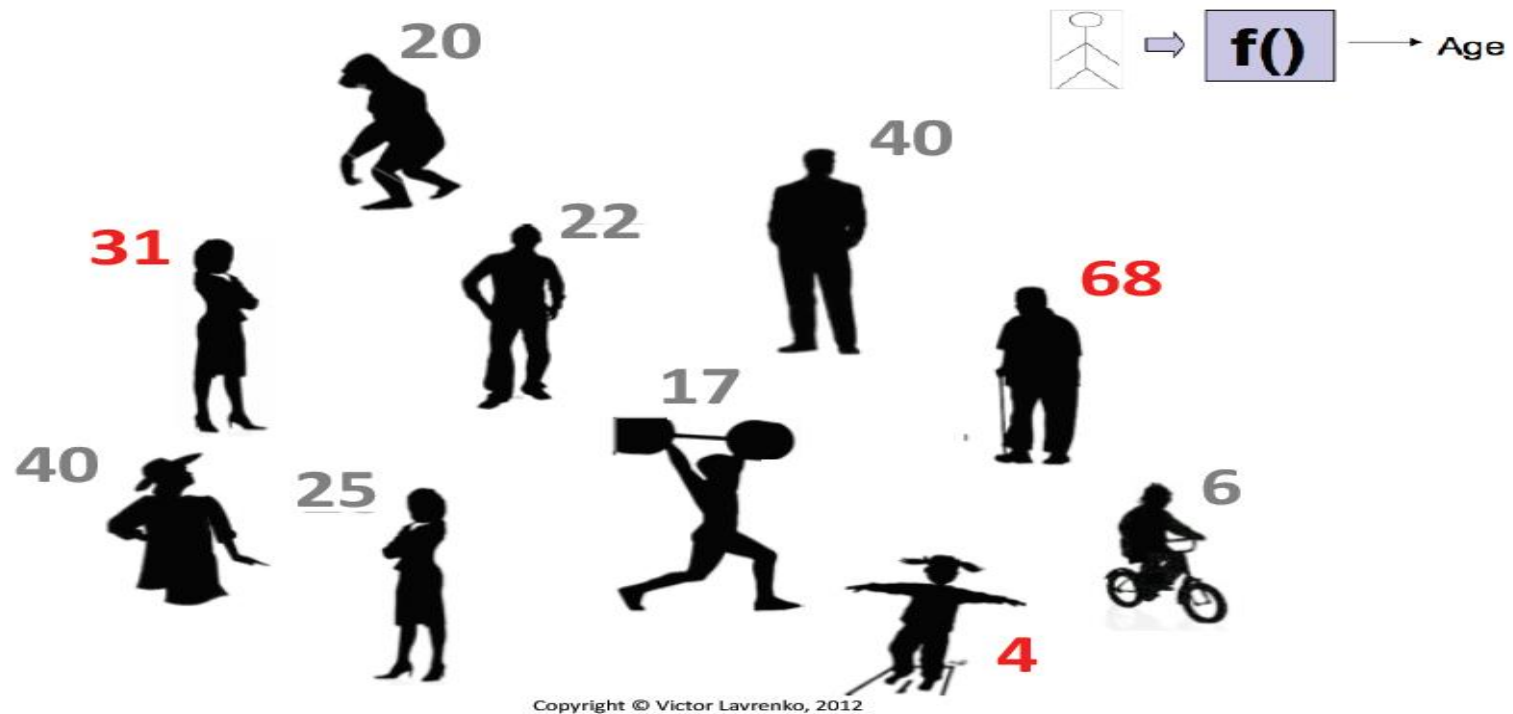
Machine Learning – Supervised Learning

Machine Learning – Supervised - Classification



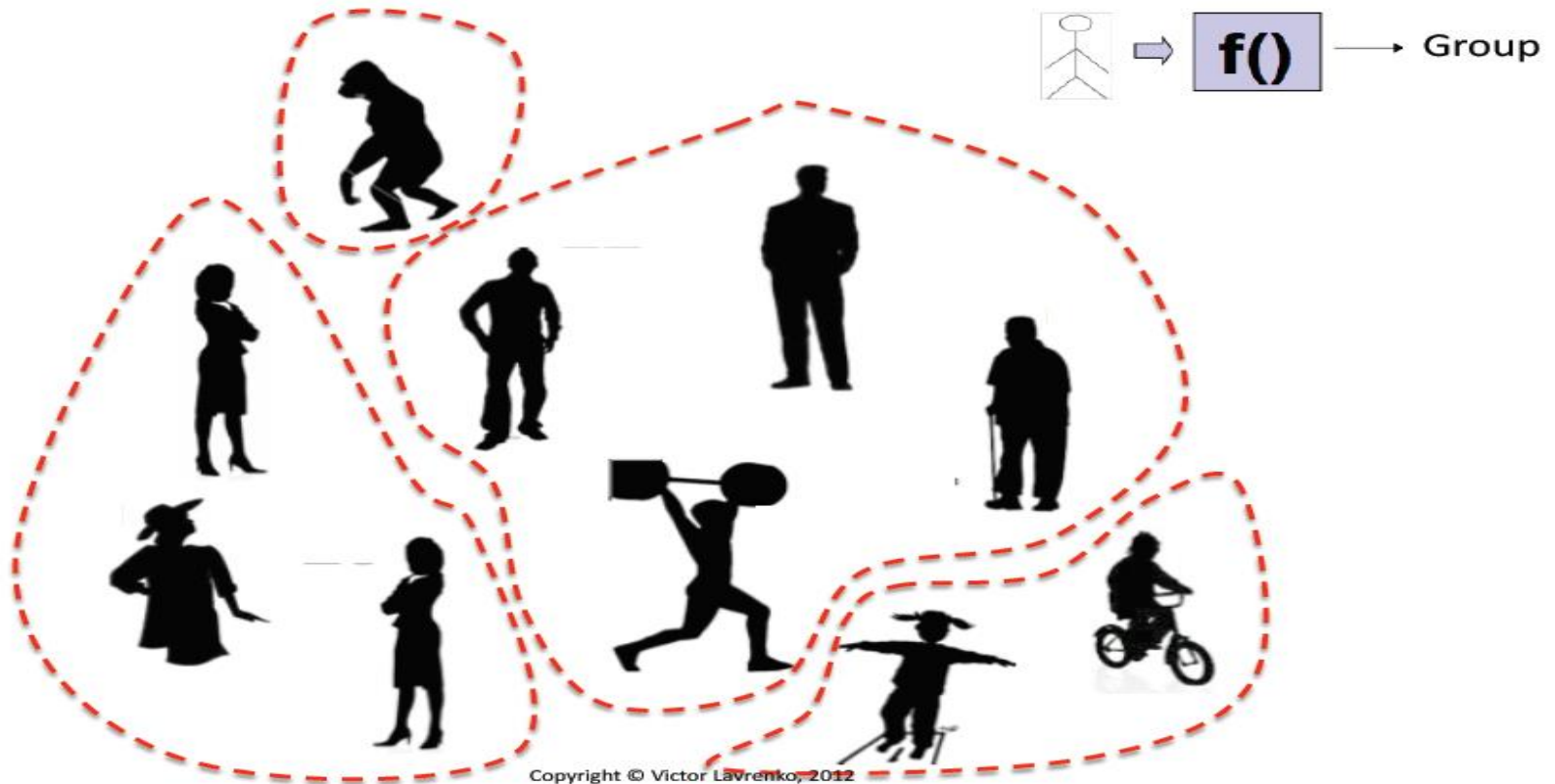
Machine Learning – Supervised Learning

Machine Learning – Supervised - Regression



Machine Learning – UnSupervised Learning

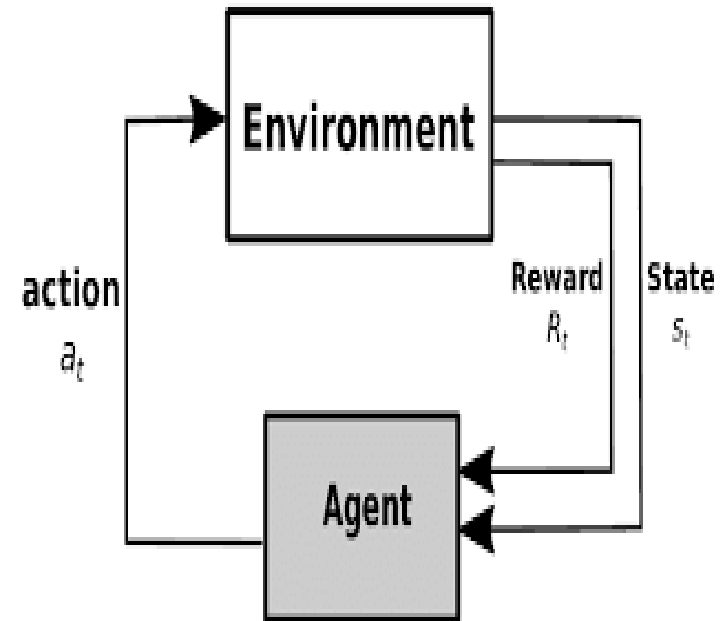
Machine Learning – UnSupervised - Clustering



Machine Learning - Reinforcement learning

Machine Learning – Reinforcement learning:

- Reinforcement learning is a sub area of Machine learning
- A reinforcement learning algorithm, or agent, learns by interacting with its environment
- The agent receives rewards by performing correctly and penalties for performing incorrectly
- The agent learns without intervention from a human by maximizing its reward and minimizing its penalty



Machine Learning - Reinforcement learning

Machine Learning – Reinforcement learning:

- Reinforcement learning will happen by making decisions sequentially. The output depends on the state of the current input and the next input depends on the output of the previous input.

Example:

- ❖ AI Games
- ❖ Chess
- ❖ Robot in a maze

Thank You